AKSHIT KUMAR

Updated: July 2026

Department of Management, University of Toronto Scarborough
Rotman School of Management (cross-appointed)
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RESEARCH INTERESTS

I am broadly interested in the operations of online platforms and marketplaces, with a particular focus on AI agents and recommendation systems.

ACADEMIC APPOINTMENTS

July 2026 - Present — **Assistant Professor, Operations Management**
Department of Management, University of Toronto Scarborough
Rotman School of Management (cross-appointed)

July 2025 - June 2026 — **Postdoctoral Researcher**
Center for Algorithms, Data and Market Design, Yale University

EDUCATION

Sep 2020 - May 2025 — **Ph.D., Operations Management**
Columbia University, Graduate School of Business
Advisors: Prof. Omar Besbes and Prof. Yash Kanoria

Jul 2018 - May 2020 — **M.S., Electrical and Computer Engineering**
University of Michigan
Advisor: Prof. Vijay Subramanian

Aug 2014 - May 2018 — **B.Tech., Electrical Engineering**
Indian Institute of Technology Madras
Advisor: Prof. Rahul Vaze

PUBLICATIONS

Working Papers

1. A. Kumar. Feature-Based Dynamic Matching in the Dark. [Link]
2. A. Kumar, V. Manshadi, A. Tumu. How Much Should a Conversational Recommender System Converse? [Link]
3. A. Allouah, O. Besbes, J. Figueroa, Y. Kanoria, A. Kumar. What Is Your AI Agent Buying? Evaluation, Biases, Model Dependence, & Emerging Implications of Agentic E-Commerce. [Link]
4. O. Besbes, Y. Kanoria, A. Kumar. Impact of Rankings and Personalized Recommendations in Marketplaces. [Link]

Journal Papers

1. Y. Chen, Y. Kanoria, A. Kumar, W. Zhang. Feature-Based Dynamic Matching. *Operations Research*, 74(2):788-803, 2026. [Link]
   - Third Place, 2025 Michael H. Rothkopf Junior Researcher Paper Prize.
   - First Place, 2024 Jeff McGill RMP Best Student Paper Prize.

2. O. Besbes, Y. Kanoria, A. Kumar. Dynamic Resource Allocation: Algorithmic Design Principles and Spectrum of Achievable Performances. *Operations Research*, 73(3):1273-1288, 2025. [Link]
   - Second Place, 2024 Michael H. Rothkopf Junior Researcher Paper Prize.
   - Finalist, 2023 INFORMS George Nicholson Student Paper Competition.
   - Finalist, 2023 Jeff McGill RMP Best Student Paper Prize.

Peer Reviewed Conference Proceedings

1. A. Kumar, V. Manshadi, A. Tumu. How Much Should a Conversational Recommender System Converse? *EC'26: ACM Conference on Economics and Computation*. [Link]

2. A. Allouah, O. Besbes, J. Figueroa, Y. Kanoria, A. Kumar. What Is Your AI Agent Buying? Evaluation, Biases, Model Dependence, & Emerging Implications of Agentic E-Commerce. *WWW'26: ACM Web Conference*. [Link]

3. O. Besbes, Y. Kanoria, A. Kumar. Impact of Rankings and Personalized Recommendations in Marketplaces. *EC'25: ACM Conference on Economics and Computation*. [Link]

4. A. Kumar, T. Peng, Y. Wu, A. Zeevi. Performance of LLMs on Stochastic Modeling Operations Research Problems: From Theory to Practice. *WSC'25: Proceedings of the 2025 Winter Simulation Conference*. [Link]

5. O. Besbes, Y. Kanoria, A. Kumar. The Fault in Our Recommendations: On the Perils of Optimizing the Measurable. *RecSys'24: ACM Conference on Recommender Systems*. [Link]

6. A. Kumar, P. Hegde, R. Vaze, A. Alloum, C. Adjih. Breaking the Unit Throughput Barrier in Random Access Protocol Based Distributed Systems. *NCC'23: 29th National Conference on Communications*. [Link]

7. Y. Chen, Y. Kanoria, A. Kumar, W. Zhang. Feature-Based Dynamic Matching. *EC'23: ACM Conference on Economics and Computation*. [Link]

8. O. Besbes, Y. Kanoria, A. Kumar. The Multi-secretary Problem with Many Types. *EC'22: ACM Conference on Economics and Computation*. [Link]

9. A. Jain, Z. Kapetanovic, A. Kumar, V. N. Swamy, R. Patil, D. Vasisht, R. Sharma, M. Swaminathan, R. Chandra, A. Badam, G. Ranade, S. Sinha, A. U. Nambi. Low-cost aerial imaging for small holder farmers. *COMPASS'19: Proceedings of the 2nd ACM SIGCAS Conference on Computing and Sustainable Societies*. [Link]
   - Best Paper Award at COMPASS'19.

10. P. Hegde, A. Kumar, R. Vaze. Speed scaling under QoS constraints with finite buffer. *WiOpt'18: 16th International Symposium on Modeling and Optimization in Mobile, Ad Hoc, and Wireless Networks*. [Link]

AWARDS

2025	Third Place, Michael H. Rothkopf Junior Researcher Paper Prize
2024	First Place, Jeff McGill RMP Best Student Paper Prize
2024	Second Place, Michael H. Rothkopf Junior Researcher Paper Prize
2024	Rising Star, Stanford Management Science and Engineering
2023	Finalist, INFORMS George Nicholson Student Paper Competition
2023	Finalist, Jeff McGill RMP Best Student Paper Prize
2023 - 2024	Deming Doctoral Fellowship, Columbia Business School
2023	Narula Doctoral Fellowship, Columbia Business School
2019	Best Paper Award, COMPASS'19
2016	Industry Category Winner at Microsoft Global Hackathon
2014	Kishore Vaigyanik Protsahan Yojana (KVPY) Fellowship
2011	National Talent Search Examination (NTSE) scholarship

PATENTS

1. R. Chandra, M. Swaminathan, V. N. Swamy, Z. Kapetanovic, D. Vasisht, A. Kumar, A. Badam, G. Ranade, S. Sinha, R. Patil. Low-cost, Long-term Aerial Imagery. [Link]
2. R. Chandra, M. Swaminathan, V. N. Swamy, Z. Kapetanovic, D. Vasisht, A. Kumar, A. Mehra, A. Gupta, S. Sinha, R. Patil. Aerial Imaging of a Region Using Above Ground Aerial Camera Platform. [Link]

INDUSTRY INTERNSHIPS

Jun 2023 - Sep 2023	**Amazon SCOT**, Bellevue, WA Developed multi-objective optimization algorithms for order fulfillment.
May 2018 - Aug 2018	**Nokia Bell Labs**, Paris, France Developed decoding schemes for distributed wireless systems.
Jun 2016 - Aug 2016	**Microsoft Research**, Bangalore, India Built low-cost precision-agriculture tools for smallholder farmers.

PROFESSIONAL SERVICE

- **Journal Reviewer:** Mathematics of Operations Research, Operations Research, Management Science, Manufacturing and Service Operations Management, Marketing Science, Information Systems Research
- **Program Committee Member:** EC 2026
- **Conference Reviewer/Sub-Reviewer:** MSOM Main Conference (2025, 2026)
- Co-organizer for the 2024 NYC Operations Day PhD Colloquium.
- DRO Internal Seminar series co-organizer (2022-2024).
- DRO PhD student representative (2022-2024).

PRESENTATIONS

What Is Your AI Agent Buying? Evaluation, Biases, Model Dependence, and Emerging Implications for Agentic E-Commerce

Workshop on Unstructured Data and Language Models, Ann Arbor, MI	Jul 2026
ACM Web Conference (Poster), Online	Jul 2026
ESIF Economics and AI+ML Meeting, Ithaca, NY	Jun 2026
Behavioral Operations Conference, Tempe, AZ	May 2026
Business Implications of Generative AI (Big.AI MIT), Boston, MA	Apr 2026
Workshop on Information Systems and Economics (WISE), Nashville, TN	Dec 2025
MIT Conference on Digital Experimentation (CODE), Boston, MA	Nov 2025
Consumer Reports, New York, NY	Nov 2025
Rotman Young Scholar Seminar, Online	Nov 2025
Yale Computer Science and Economics Day, New Haven, CT	Sep 2025
Workshop on AI and the Future of Collaborative Innovation, New York, NY	Sep 2025

Impact of Rankings and Personalized Recommendations in Marketplaces

INFORMS Annual Meeting, Atlanta, GA	Oct 2025
Revenue Management and Pricing (RMP) Conference, New York, NY	Jul 2025
EC'25: ACM Conference on Economics and Computation, Stanford, CA	Jul 2025

Dynamic Resource Allocation: Theory and Practice

From Matching to Markets Workshop, Corsica, France	May 2025
Industrial and Systems Engineering, University of Minnesota, Minneapolis, MN	Jan 2025
Mendoza College of Business, University of Notre Dame, Notre Dame, IN	Jan 2025
Yale School of Management, New Haven, CT	Jan 2025
Stanford Graduate School of Business, Stanford, CA	Jan 2025
Cornell ORIE, Ithaca, NY	Dec 2024
Cornell Tech ORIE, New York, NY	Dec 2024
Imperial College Business School, London, England	Dec 2024
Rotman School of Management, University of Toronto, Toronto, Canada	Dec 2024
Indian School of Business, Hyderabad, India	Nov 2024
NUS Business School, Singapore	Nov 2024
NYU Stern School of Business, New York, NY	Nov 2024
Wharton School of the University of Pennsylvania, Philadelphia, PA	Nov 2024

The Fault in Our Recommendations: On the Perils of Optimizing the Measurable

Yale FDS Workshop on Social Algorithms (Poster)	Oct 2025
Netflix ML Seminar, Online	Nov 2024
RecSys'24: ACM Conference on Recommender Systems, Bari, Italy	Oct 2024

Dynamic Resource Allocation: Algorithmic Design Principles and Spectrum of Achievable Performances

Workshop on Operations Research and Data Science (WORDS), Duke Fuqua, Durham, NC	Nov 2024
Operations Rookiepalooza, Kellogg School of Management, Chicago, IL	Oct 2024
INFORMS Annual Meeting, Seattle, WA	Oct 2024
Cornell Young Research Workshop, Ithaca, NY	Oct 2024
Purdue Operations Conference, West Lafayette, IN	Aug 2024
Industrial Engineering and Operations Research (IEOR) Seminar, IIT Bombay, Mumbai, India	Aug 2024
International Symposium on Mathematical Programming (ISMP), Montreal, Canada	Jul 2024
Revenue Management and Pricing (RMP) Conference, Los Angeles, CA	Jul 2024
MS&E Rising Stars Workshop, Stanford, CA	Apr 2024

INFORMS Annual Meeting, Phoenix, AZ — Oct 2023
Fulfillment Optimization Research Series, Amazon — Aug 2023
Tata Institute of Fundamental Research (TIFR), Mumbai, India — May 2023

Feature-Based Dynamic Matching

INFORMS Annual Meeting, Atlanta, GA — Oct 2025
INFORMS Annual Meeting, Seattle, WA — Oct 2024
Manufacturing and Service Operations Management (MSOM) Conference, Minneapolis, MN — Jul 2024
Marketplace Innovation Workshop, Online — May 2024
Revenue Management and Pricing (RMP) Conference, London, England — Jun 2023

A PROOF: Approximately PaReto Optimal Order Fulfillment

Deming Doctoral Fellowship Seminar, Columbia Business School — May 2024

The multi-secretary problem with many types

INFORMS Annual Meeting, Indianapolis, IN — Oct 2022
EC'22: ACM Conference on Economics and Computation, Boulder, CO — Jul 2022
Manufacturing and Service Operations Management (MSOM) Conference, Munich, Germany — Jun 2022
Revenue Management and Pricing (RMP) Conference, Online — Jun 2022

SKILLS

Programming: Python, C/C++, MATLAB, HTML/CSS
Tools: Git, LaTeX